



ORACLE

MySQL Implementation Essentials Bootcamp

Oracle Cloud Infrastructure

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2023



Program Agenda

- Introduction
- Oracle MySQL Enterprise
- **Oracle Cloud Infrastructure overview**
- Installation and Architecture
- Database Design
- Security
- Backup
- Replication
- High Availability
- MySQL Enterprise Monitor
- HeatWave

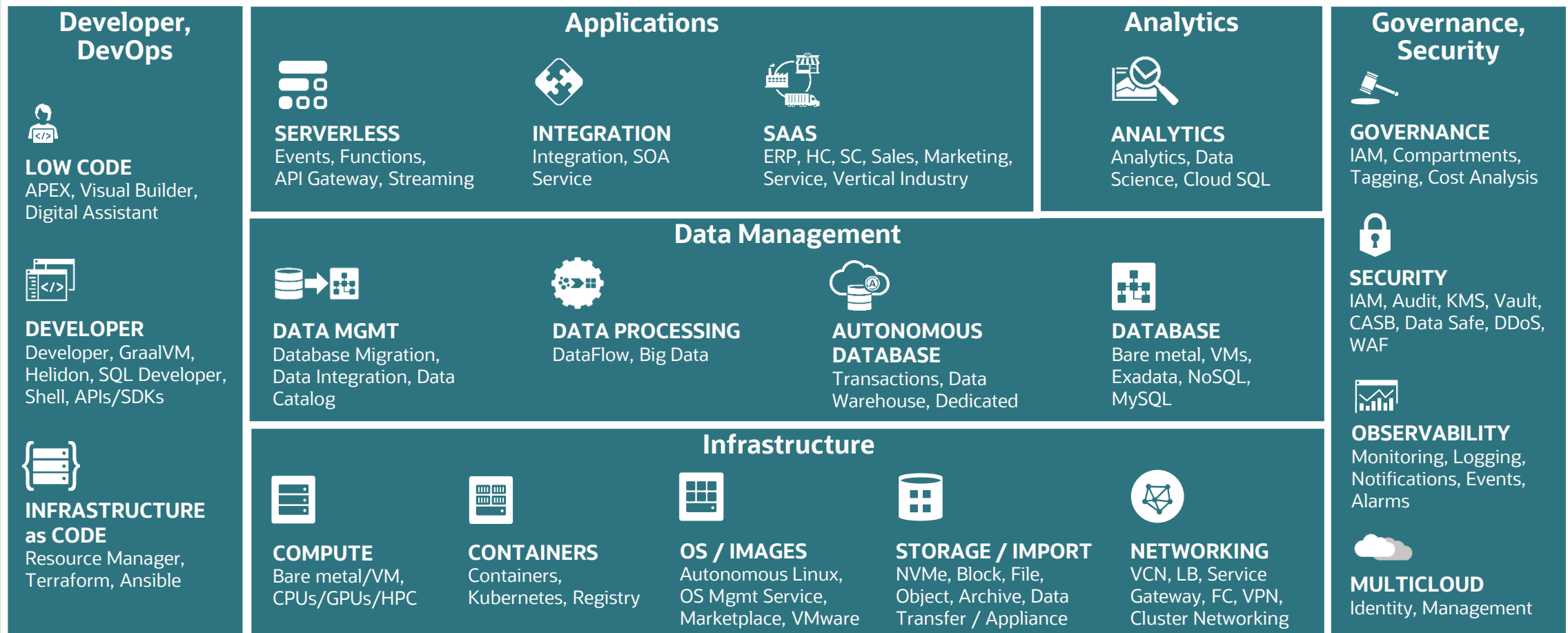
Stay updated: others will be added!

Oracle Cloud Infrastructure

- Commercial
- Commercial Planned
- Government
- Microsoft Interconnect Azure



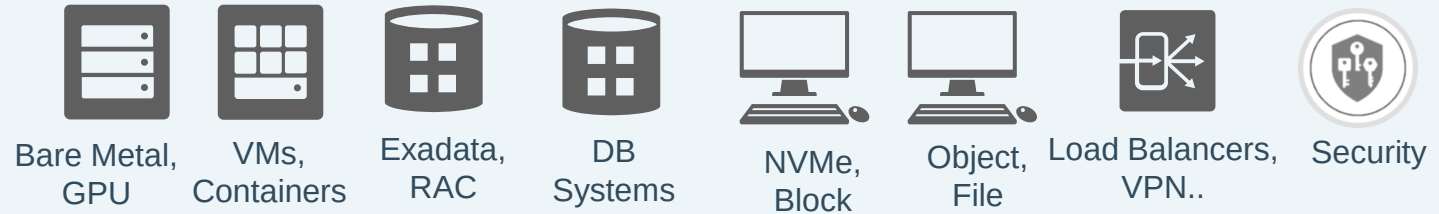
Oracle Cloud Infrastructure: Complete cloud services



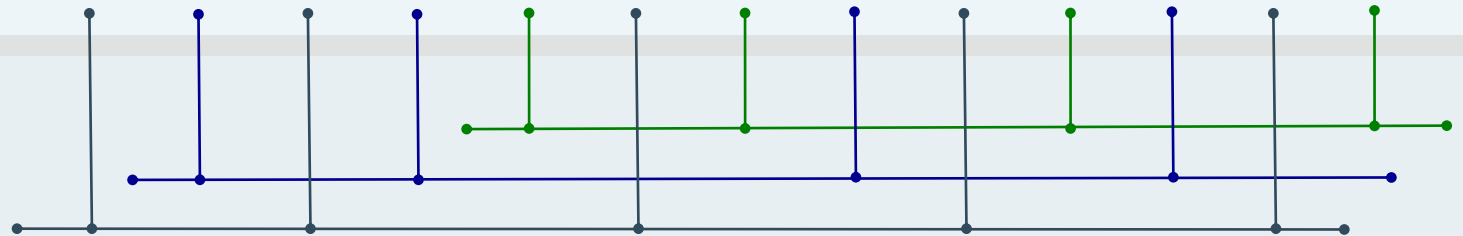
PUBLIC AND GOVERNMENT REGIONS / CLOUD AT CUSTOMER / AZURE

Oracle Cloud Infrastructure Services

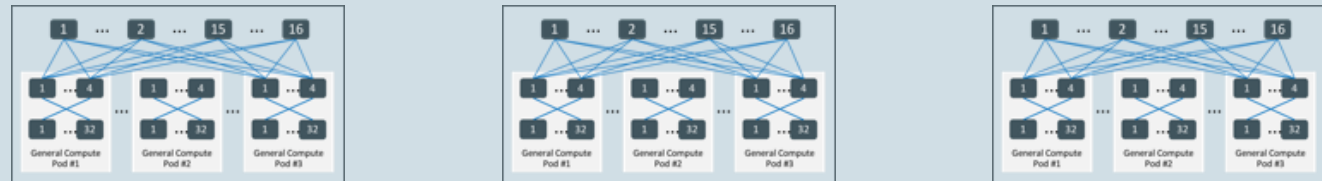
**COMPUTE, STORAGE,
DATABASE, LBs, Security...**



VIRTUAL NETWORK



PHYSICAL NETWORK



REGION

DATACENTERS

Availability
Domain 1



Availability
Domain 2

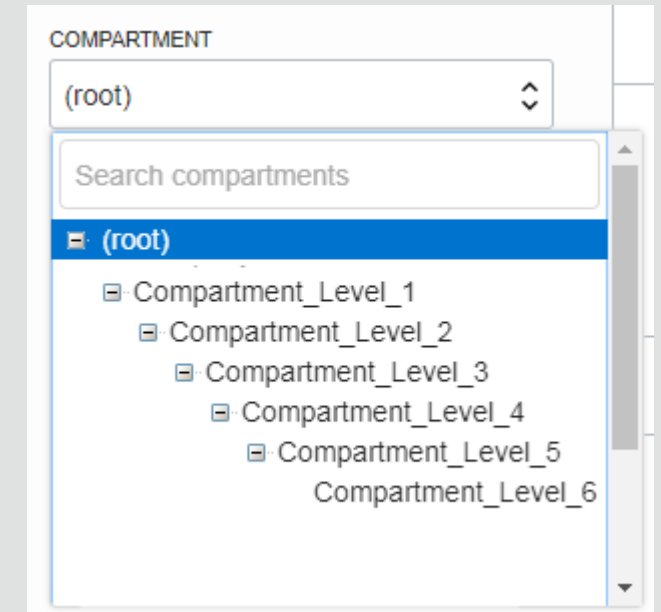


Availability
Domain 3



Compartment

- A collection of related resources that can be accessed only by groups that have been given permission by an administrator in your organization
- First compartment is "(root)"
- Compartments have a hierarchy structure
 - sub-compartments inherit all access permissions from the compartments higher up in the hierarchy



<https://docs.oracle.com/en-us/iaas/Content/Identity/Tasks/managingcompartments.htm>

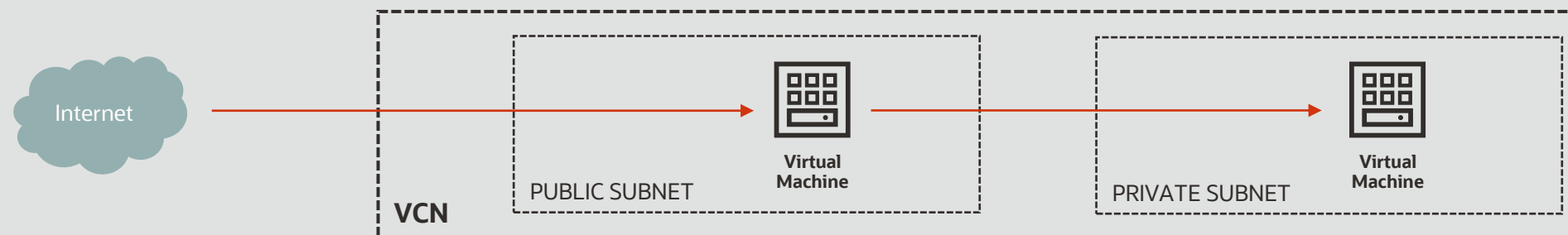
Virtual Cloud Network

- A VCN is a customizable private network in Oracle Cloud Infrastructure
- Just like a traditional data center network, a VCN provides you with complete control over your network environment
 - This includes assigning your own private IP address space, creating subnets, creating route tables, and configuring stateful firewalls
- A single tenancy (an Oracle Cloud Infrastructure account) can have multiple VCNs, thereby providing grouping and isolation of related resources
 - For example, you might use multiple VCNs to separate the resources in different departments in your company

<https://docs.oracle.com/en-us/iaas/Content/Network/Concepts/overview.htm>

Subnets

- **Public subnet**
 - A subnet in which instances are allowed to have public IP addresses. When you launch an instance in a public subnet, you specify whether the instance should have a public IP address
- **Private subnet**
 - A subnet in which instances are not allowed to have public IP addresses
 - Each instance has at least one primary private IP address
- **Regional Subnet**
 - A subnet that spans all the availability domains in the region. Oracle recommends using regional subnets because they are more flexible and make it easier to implement failover across availability domains



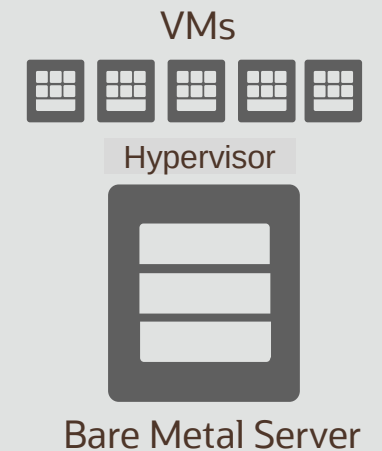
Example: Ingress Rule

- Ingress Rules apply to incoming connections
- Egress Rules apply to outgoing connections
- In the example below we create a rule that allow connections from everywhere (e.g. internet) to the destinations in the subnets on the port 80

SOURCE TYPE	SOURCE CIDR	IP PROTOCOL	SOURCE PORT RANGE (OPTIONAL)	DESTINATION PORT RANGE (OPTIONAL)
CIDR	0.0.0.0/0 Specified IP addresses: 0.0.0.0-255.255.255.255 (4,294,967,296 IP addresses)	TCP (more information)	All Examples: 80, 20-22 or All (more information)	80 Examples: 80, 20-22 or All (more information)

Compute Instance

- Oracle Cloud **Infrastructure Compute** lets you provision and manage compute hosts, known as instances
- After you create an instance, you can access it securely from your computer, restart it, attach and detach volumes, and terminate it when you're done with it
 - Any changes made to the instance's local drives are lost when you terminate it
 - Any saved changes to volumes attached to the instance are retained
- Oracle Cloud Infrastructure offers both bare metal and virtual machine instances:
 - Bare Metal: A bare metal compute instance gives you dedicated physical server access for highest performance and strong isolation
 - **Virtual Machine**: A virtual machine (VM) is an independent computing environment that runs on top of physical bare metal hardware. The virtualization makes it possible to run multiple VMs that are isolated from each other
- VMs are ideal for running applications that do not require the performance and resources (CPU, memory, network bandwidth, storage) of an entire physical machine



<https://docs.oracle.com/en-us/iaas/Content/Compute/home.htm>



Compute

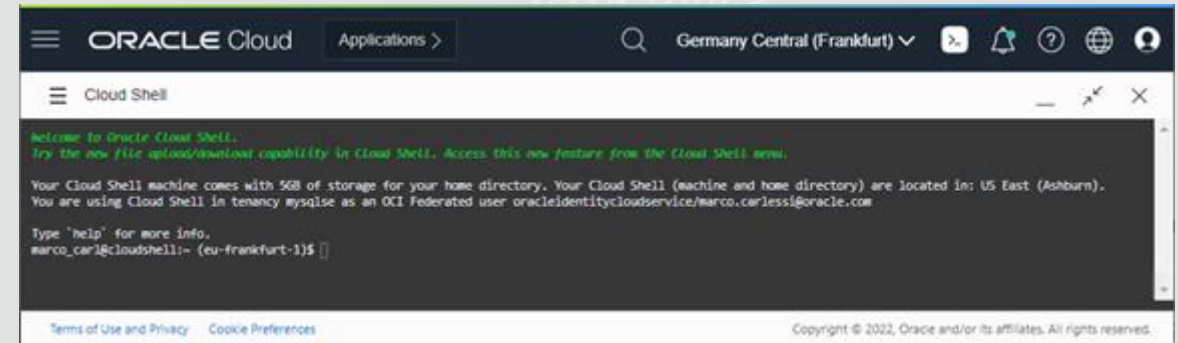
Compute services
for any enterprise
use case

Bare Metal	VMs	Containers	Functions
Instance isolation Highest IOPS High throughput Low latency	Security- hardened hypervisor Flexible sizing Dense IO and Dedicated host options	Bare metal performance Self-healing clusters	Pay only for usage Serverless Container-native Open source
AMD EPYC		Intel Xeon	NVIDIA GPUs
Local Attached Storage		Remote Attached Storage	
NVMe SSDs Up to 51.2 TB Millions of IOPS		NVMe Block Volumes up to 1PB 32 TB / volume 75 IOPS / GB	

OCI Tools

- **OCID (Oracle Cloud ID)**
 - An Oracle-assigned unique ID for a cloud resource
 - Used to address an OCI resource
 - OCID syntax: `ocid1.<RESOURCE TYPE>.<REALM>.[<REGION>][.<FUTURE USE>].<UNIQUE ID>`
 - Example: `ocid1.instance.oc1.phx.abuw4ljrlsfjqw6vzzxb43vyypt4pkodawglp3wqxjqofakrwwou52gb6s5a`
- **OCI Command Line Interface (OCI CLI)**
 - a small-footprint tool that you can use on its own or with the Console to complete OCI tasks
 - The CLI provides the same core functionality as the Console, plus additional commands. Some of these, such as the ability to run scripts, extend Console functionality
 - Examples to stop & start compute instances

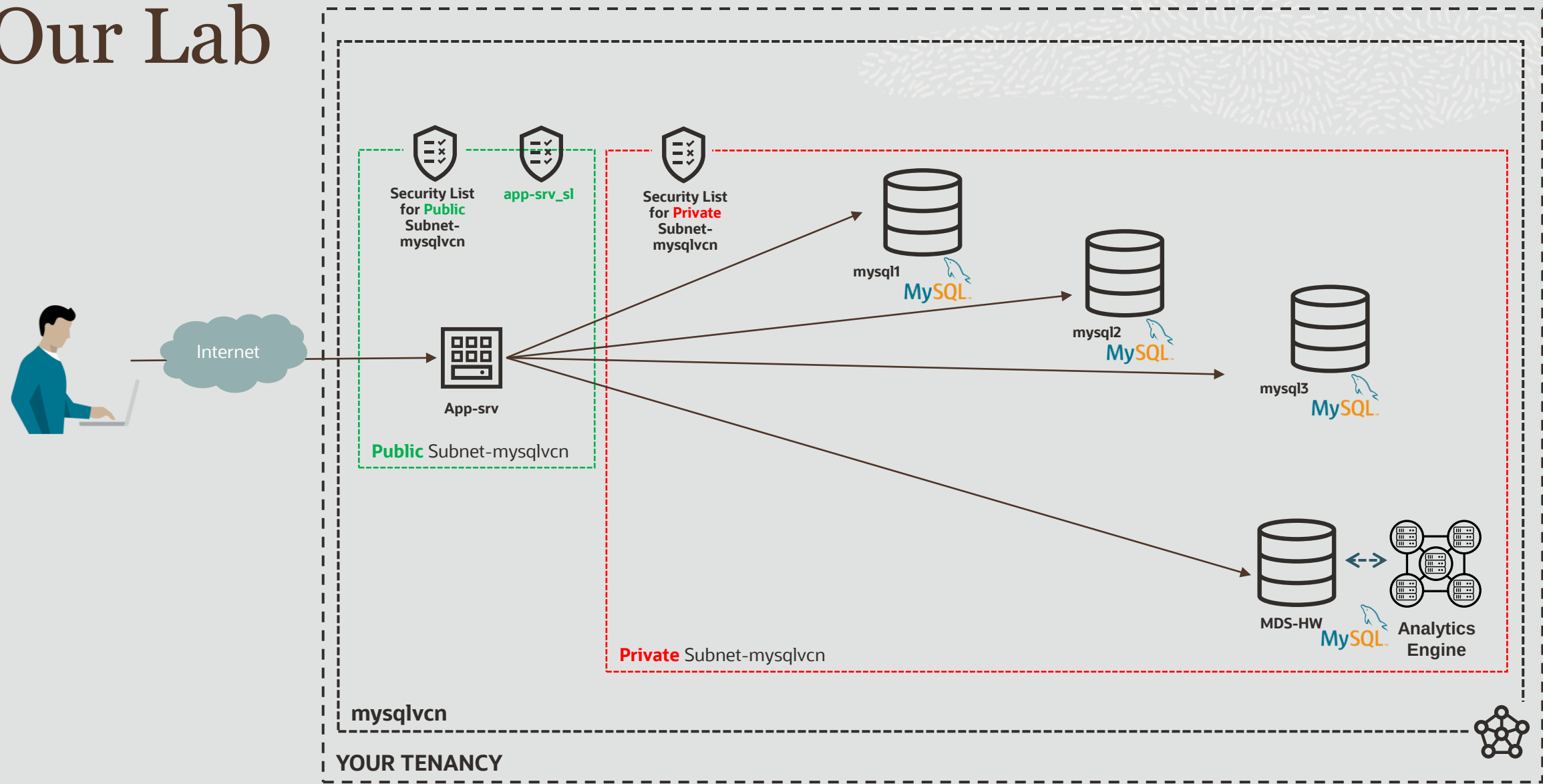
```
$> oci compute instance action --action START --instance-id 'ocid1.instance.oc1...'
$> oci compute instance action --action SOFTSTOP --instance-id 'ocid1.instance.oc1...'
```
- **Cloud Shell: a web browser-based terminal accessible from the Oracle Cloud Console**
 - free to use (within monthly tenancy limits)
 - provides access to a Linux shell, with a pre-authenticated OCI CLI, a pre-authenticated Ansible installation, and other useful tools

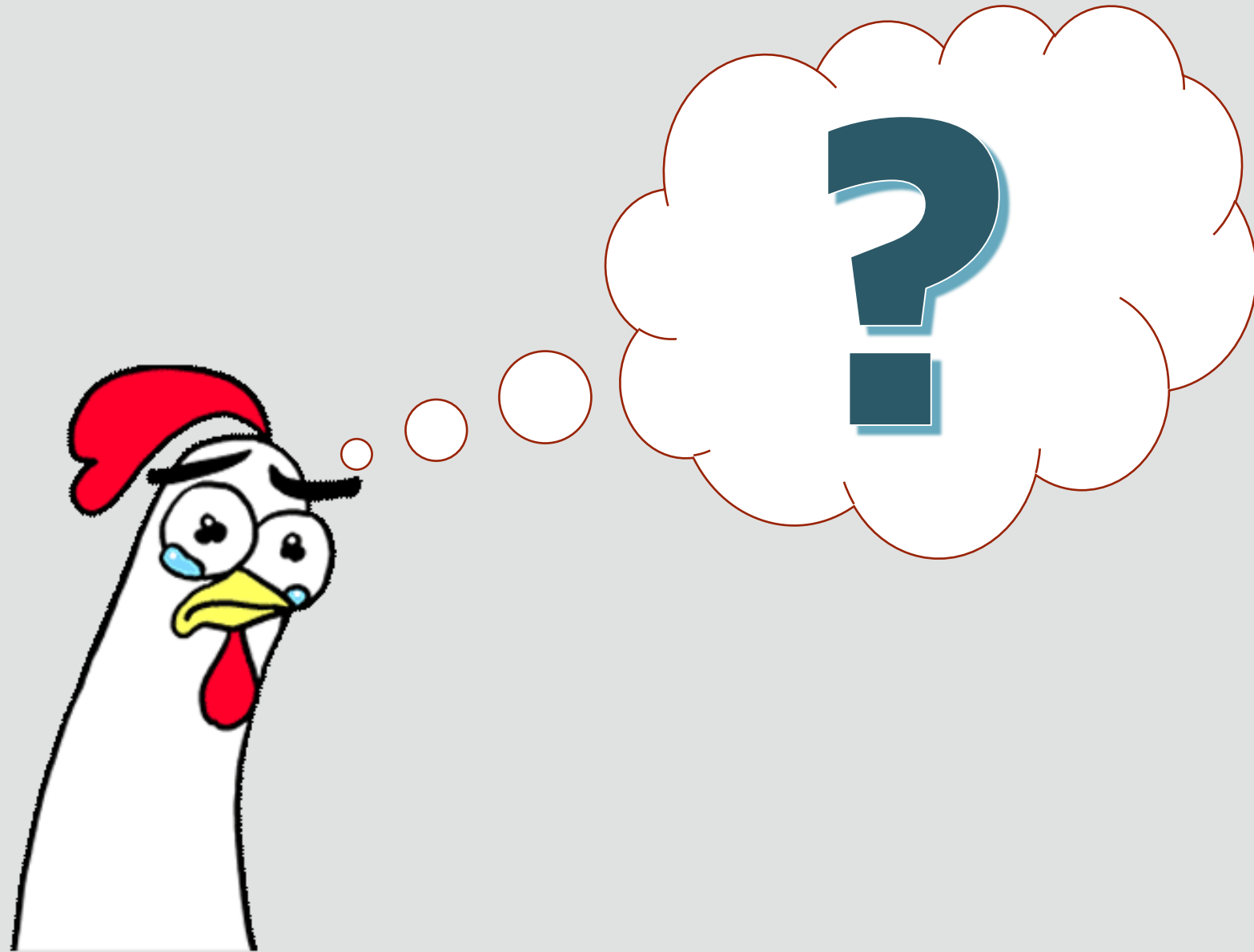


OCI Glossary

Tenancy	A secure and isolated partition where you can create, organize, and administer your cloud resources.
Region	A collection of availability domains located in a single geographic location
Availability Domain	One or more isolated, fault-tolerant Oracle data centers that host cloud resources such as instances, volumes, and subnets
Fault Domain	A logical grouping of hardware and infrastructure within an availability domain. Fault domains isolate resources during hardware failure or unexpected software changes
Compartment	A collection of related resources that can be accessed only by groups that have been given permission by an administrator in your organization
Compute	Compute hosts, known as instances
Shape	A template that determines the number of CPUs, amount of memory, and other resources that are allocated to a compute instance
VCN (Virtual Cloud Network)	A virtual version of a traditional network — including CIDRs, subnets, route tables, and gateways — on which your instance runs
Subnet	Subnets separate your network into multiple smaller and distinct networks
Security list	A list of security rules that apply to all resources in any subnet that uses the security list
OCID	An Oracle-assigned unique ID for a cloud resource, used to address an OCI resource

Our Lab





DEMO

- OCI infrastructure



Hands-On Labs



1a. Download material and test lab connection



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